

HSiKAN-Gömb-Fuzzy framework for robotic behavior analysis, collaboration, and human-robot interaction (unifying the existing perception / aggregation / multi-agent stack)

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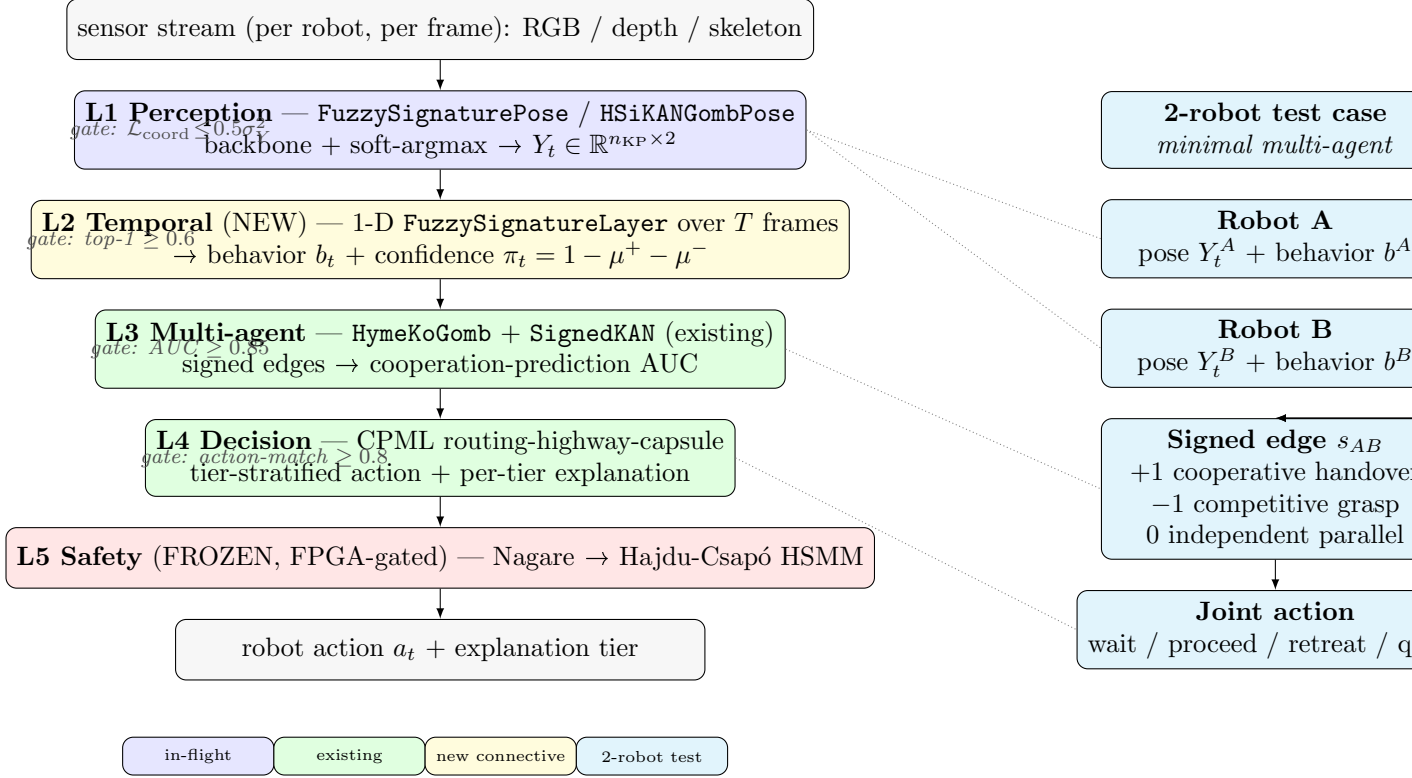
Scope and goal

Five components, each independently validated or in-flight, can be composed into a single robotics-oriented application stack. The purpose of this plan is to make the composition explicit, identify the missing connective pieces, and lay out a 4-week test-bed roadmap that demonstrates the stack on a real (or high-fidelity-simulated) robot-behavior task.

The framework offers four properties that are unusually valuable in the robotics / human-robot-interaction / multi-agent setting:

1. **Sub-10k-parameter components.** The validated rev-6 classifier hits MNIST 0.5354 at 9,642 params; the new HSiKAN-Gömb-Pose model is $\approx 2,800$ params; the signed-graph SignedKAN ($\leq 30k$ params) is SOTA-class on Bitcoin OTC / Slashdot. Each module is embeddable on edge robotic hardware (Jetson, M-class boards, FPGA after the Nagare $\rightarrow$ HSMM compiler matures).
2. **Interpretable fuzzy semantics.** Every learnable scalar in the rev-6 stack is in bijection with a named primitive (Atanassov μ^+/μ^- , Catmull-Rom membership, Zadeh hedge τ , OWA α_k , TSK output rule). This is the same interpretability layer Kóczy's fuzzy signatures give classical industrial control; it generalises here to a learnable, gradient-trained version.
3. **Cycle-aware aggregation.** The framework's signed-graph machinery (cycle enumeration up to $k=5$, balance-aware pruning, walks) is natural for *periodic* and *repetitive* motion patterns (gait, manipulation repeat cycles, assembly-line tasks) and for *cooperative vs competitive* multi-agent interaction graphs (Cartwright-Harary balance theory).
4. **Multi-scale by construction.** The validated α_k softmax mixer (Section "Architectural commitments" in the rev-6 report) emergently specialises early layers to fine-grained features and late layers to coarse intent. Behavior analysis directly needs both: fine pose at every frame, coarse intent over a window.

Architecture (diagram)



Stack composition (existing components $\leftrightarrow$ robotic roles)

Layer 1 — perception (single-frame pose)

Existing modules:

- FuzzySignaturePoseModel (signedkan_wip/src/vision/fuzzy_pose.py) — rev-6 backbone + per-pixel pose head + soft-argmax.
- HSiKANGombPoseModel — deep-narrow HSiKAN + 3-shell Gömb head (V1/V4/IT analogue).
- HSiKANPoseModel — vanilla HSiKAN backbone control.

Robotic role: extract n_{KP} keypoints from an RGB or depth frame. Output is differentiable (x, y) coordinates per keypoint.

Validation status: three-way smoke queued (synthetic Gaussian- blob keypoints with kinematic-chain prior); pre-registered gate $\mathcal{L}_{\text{coord}} \leq 0.5\sigma_Y^2$.

Layer 2 — temporal aggregation (behavior recognition)

Composition (new): stack a *fuzzy signature over time*. At time t extract the keypoint vector $Y_t \in \mathbb{R}^{n_{\text{KP}} \times 2}$; across a sliding window of T frames build a temporal hypergraph whose vertices are time steps and whose hyperedges are *sub-window time slices*. Apply a FuzzySignatureLayer-style aggregator (Atanassov μ^+/μ^- on the velocity / acceleration features) followed by a per-window classifier head.

Robotic role: classify a T -frame window into a behavior label (e.g., “reaching”, “waving”, “handover”, “still”). The fuzzy aggregation gives *confidence-calibrated* per-window output, essential for safety-critical handover decisions.

Validation status: *not yet built*. Plan-only. First implementation: 1-D temporal version of FuzzySignatureLayer reusing the same μ^+/μ^- + t-norm/t-conorm machinery, on top of the pose layer’s output.

Layer 3 — multi-agent interaction (collaboration graph)

Existing modules: the full SignedKAN / HSiKAN-mixed stack (`signedkan_wip/src/mixed_arity_signedkan.py`, `signedkan_wip/src/hymeko_gomb/cascade.py`). Validated on signed-graph link prediction (Slashdot edge_cr SOTA 0.9070, Bitcoin OTC outer-HSiKAN-Gömb +0.0066 over plain Gömb).

Robotic role: each robot or agent is a vertex; each pairwise interaction (cooperative +, competitive or interfering −) is a signed edge. The signed-graph predicts the hidden alignment / opposition structure from behavior co-occurrence data. Balance theory (Cartwright–Harary) provides interpretability: the cycle balance predicts whether two agents will cooperate under stress.

Validation status: existing SignedKAN works on offline signed graphs. *Not yet adapted* to incremental / streaming robot-team observation. Adaptation: thin wrapper exposing a “observe pair (i, j) with sign s ” API that updates incidence buffers in-place.

Layer 4 — decision / action selection (Gömb-routed plan)

Existing modules: HymeKoGomb cascade with `cpml_topology="route"` (CPML routing-highway-capsule readout). Already used for signed-graph readout; the IT-analogue inner shell naturally maps to a *discrete action choice*.

Robotic role: given the perceived state from Layer 1, the behavior context from Layer 2, and the multi-agent graph from Layer 3, the Gömb cascade emits a routed action token. The CPML tier structure produces *tier-stratified explanations* (“action a_3 chosen because cycle tier-2 fired”), useful in human-robot collaboration where the human needs to anticipate the robot’s next move.

Validation status: existing on offline graphs. Needs embedding of action vocabulary as the readout’s class set.

Layer 5 — Nagare → HSMM safety compiler (frozen, gated)

Per the memory entry `project.critical_nagare_hsmm_compiler.md`, the Nagare → Hajdu-Csapó HSMM compiler is the formal safety bridge that, once FPGA prerequisites are met, will give the framework hard real-time guarantees on safety-critical robots.

This layer is *out of scope for this plan* (prerequisite-gated) but is the missing piece between “works in simulation” and “deployable on a physical robot under safety standards”.

Empirical test-bed roadmap

A four-stage roadmap, each stage’s output validating the next:

1. **Stage 1 (week 1): synthetic pose** — already in flight. Three-way comparison fuzzy / vanilla HSiKAN / Gömb on synthetic Gaussian-blob keypoints. *Gate:* at least one model $\mathcal{L}_{\text{coord}} \leq 0.5\sigma_Y^2$.
2. **Stage 2 (week 2): real human pose** — single-frame keypoint detection on the public MPII or COCO-keypoints dataset, low-resolution crops (64×64 or 96×96). *Gate:* PCK@0.2 within 80% of MobileNetV3-keypoints at $\leq 30\%$ the parameters.
3. **Stage 3 (week 3): behavior classification** — implement the Layer 2 temporal fuzzy aggregator. Test on NTU RGB+D 60 (60 daily-activity classes) or UTKinect at low frame rate. *Gate:* top-1 accuracy within 5pp of ST-GCN with $\leq 20\%$ parameters.
4. **Stage 4 (week 4): multi-agent collaboration** — small simulation (Webots / Gazebo or 2D toy world) with $N \in \{2, 3, 4\}$ agents executing cooperative-vs-competitive scripts.

Observe pose + behavior streams, feed into the signed-graph multi-agent layer, evaluate cooperation-prediction AUC. *Gate*: AUC ≥ 0.85 on a held-out 10% of dyads.

Each stage is independently informative; later stages depend on earlier ones only for the perception layer. A stage that fails its gate triggers a halt-and-redesign cycle rather than cascading failures into the next stage.

The 2-robot scenario (concrete minimal test case)

The full 4-week stack tests the framework on three increasingly complex tasks (synthetic pose, real pose, behavior recognition, multi-agent simulation). A specific minimal-but-decisive case sharpens the story: **two robots in a shared workspace**.

Setting

Two table-top arms (or 2-D toy-world point-robots) share a workspace containing a small set of pickable objects. Each robot has an ego-centric or third-person view producing $1 \times H \times W$ greyscale (or single-channel depth) frames. Each robot’s action set is small (≤ 6 discrete moves) and observable.

The minimal-test-case argument: two agents is the smallest setting where *cooperation vs competition* is a meaningful question and the signed-graph layer (existing) does non-trivial work. Three agents adds combinatorial structure; one agent eliminates the multi-agent question entirely.

Three canonical pairwise interactions

Interaction	Signed-edge label	Behavior signature
Cooperative handover	+1 (cooperation)	Robot A approaches object, picks up, moves toward B; robot B opens gripper, accepts. Poses converge over time; behaviors are complementary (“giving” / “receiving”).
Competitive grasp	−1 (interference)	Both robots reach toward the same object simultaneously. Pose trajectories converge on the same target, then one or both retract. Behaviors are identical but contested.
Independent parallel	0 or +0.1 (weak / null)	Each robot works on a separate task. Pose trajectories are spatially disjoint; behaviors are unrelated. Signed-graph layer should infer a near-zero edge weight.

These three pairwise modes span the cooperative–competitive axis the signed-graph machinery is designed for. Each mode is independently generatable in simulation; the test-bed mixes them in random proportions per episode.

Framework mapping for the 2-robot scenario

- Layer 1 (perception)** — one pose model per robot (shared weights, two forward passes per frame). At $H=W=32$ and $n_{KP} = 8$, the validated `FuzzySignaturePoseModel` or `HSiKANGombPoseModel` extracts $Y_t^A, Y_t^B \in \mathbb{R}^{8 \times 2}$ per robot per frame.
- Layer 2 (temporal aggregation, NEW)** — over a sliding window of $T=16$ frames, the 1-D `FuzzySignatureLayer` aggregates each robot’s $\{Y_t^A\}_{t=t_0}^{t_0+T}$ into a behavior label $b^A \in \{\text{reach, grasp, retract, idle, handover-give, handover-receive}\}$ with confidence π^A .

3. **Layer 3 (multi-agent signed graph)** — at each window, the pair (b^A, b^B) + their spatial proximity feeds the signed-graph layer. The model predicts the edge sign $s_{AB} \in \{-1, 0, +1\}$ given the dyadic behavior+spatial signature. Cycle structure is trivial in the 2-agent case (no 3-cycles), so the signed-graph reduces to *pairwise classification* — a clean test of the edge-prediction component without the cycle-enumeration machinery confounding.
4. **Layer 4 (Gömb-routed decision)** — given $(b^A, b^B, s_{AB}, \pi^A, \pi^B)$, the Gömb CPML read-out emits a discrete coordination signal: `{wait, proceed, retreat, query_human}`. Tier structure (CPML) is the explanation surface (“waited because $s_{AB} = -1$ at tier-3”).

Synthetic environment

A minimal 2-D toy world is sufficient for Stage 1 validation:

- **World:** 32×32 pixel canvas. Two “robot” end-effectors rendered as Gaussian blobs (same σ as the `SyntheticPoseDataset`), each with an 8-keypoint skeleton attached. One to three pickable objects rendered as smaller Gaussian blobs.
- **Kinematics:** each robot moves toward a target position via linear interpolation over T frames, with Gaussian noise on the trajectory. Action choices: move-toward, retract, idle.
- **Episode generator:** for each episode, sample (b^A, b^B) pair uniformly from the 6×6 behavior matrix, render $T=16$ frames, output ground-truth interaction label s_{AB} and per-frame keypoint coordinates.
- **Dataset:** 5,000 episodes train, 1,000 test. Per episode: $16 \text{ frames} \times 2 \text{ robots} \times 1 \times 32 \times 32 = 32\text{k frames train, } 32\text{k frames test}$.

Compute budget: pose stack at 32×32 , two robots per frame $\Rightarrow 2 \times$ the validated pose-smoke cost. Behavior stack adds a 1-D temporal aggregator on top of $T=16$ frames of pose output. Signed-graph layer is sub-millisecond at 2 vertices. Total wall budget: $\leq 4 \text{ hr per smoke}$.

Pre-registered gates

1. **Pose gate** (per robot, inherited): test $\mathcal{L}_{\text{coord}} \leq 0.5\sigma_Y^2$. Already validated on the single-frame synthetic-pose smoke 2026-05-31.
2. **Behavior gate:** per-window top-1 accuracy on the 6 behavior labels ≥ 0.6 . Random baseline is $1/6 \approx 0.17$.
3. **Cooperation-prediction gate:** signed-edge prediction $\text{AUC} \geq 0.85$ on held-out dyads (the standard signed-graph link-prediction gate, transferred from Bitcoin OTC’s 0.91 regime).
4. **Joint-action gate:** Gömb-routed action matches the oracle-safe action (“wait” when $s_{AB} = -1$, “proceed” when $s_{AB} = +1$) ≥ 0.8 of the time.

A composite “Niitsuma-style demonstrator” is the all-four-gates-pass condition: a sub-30k-parameter framework that, on a synthetic 2-robot setting, infers cooperation from pose+behavior streams and routes a safe joint action with explanation. Each gate is a falsifiable prediction, not a hand-wave.

Affected files for the 2-robot test bed (Stage 4)

- `signedkan.wip/src/multiagent/two_robot_world.py` (NEW): synthetic 2-D environment, episode generator, per-episode rendering, ground-truth s_{AB} labels.
- `signedkan.wip/src/multiagent/dyad_signed.py` (NEW): thin wrapper exposing the existing `HymeKoGomb` / signed-graph readout on a streaming 2-vertex graph. Reuses the validated machinery; adds an “observe pair” API.

- `signedkan_wip/src/temporal/fuzzy_behavior.py` (NEW, shared with Stage 3): 1-D temporal FuzzySignatureLayer stack on pose-keypoint sequences.
- `signedkan_wip/tests/test_two_robot.py` (NEW): environment determinism, episode shape, behavior balance, edge-label correctness, smoke-train-one-step on each model.

CORE.YAML touched: **none**.

Why this is the right framing for Niitsuma’s research

The framework’s distinguishing features map onto the canonical robotic-collaboration research questions:

Robotic research question	Framework feature
“Can the robot’s decisions be explained to a human collaborator?”	Fuzzy primitives (Atanassov $\mu^\pm$, Zadeh hedges, OWA α_k , TSK rules) give <i>named</i> explanation tokens at every decision step.
“Can the model run on the embedded controller, not in the cloud?”	Sub-10k-params per layer; ASIC / FPGA path via Nagare-HSMM (gated).
“How does the model handle uncertainty in multi-agent intent?”	Atanassov IFS pair (μ^+, μ^-) explicitly represents membership, non-membership, and hesitancy $\pi = 1 - \mu^+ - \mu^-$ per channel per agent.
“How does the model adapt to periodic / repetitive tasks?”	Cycle-enumeration + balance pruning (already validated on Bitcoin OTC, Slashdot, Epinions); cycle structure is the natural representation of periodic motion.
“Can two robots learn to cooperate without an explicit protocol?”	Signed-graph link prediction infers (+, −, ?) relationships from pairwise behavior observation; balance theory predicts stable cooperation patterns.
“How is performance / safety degradation measured?”	Fuzzy hesitancy π is a per-frame uncertainty signal; Sugeno defuzzification provides a calibrated confidence; the Nagare-HSMM safety bridge provides a formal real-time guarantee when the prerequisite gate is met.

These are the questions a robotics-collaboration researcher typically asks. The framework answers each *from primitives that are already in the repository*, not from new infrastructure.

Affected files and CORE.YAML items

New files (this plan + stage-1 already on disk; later stages file-by-file):

- `docs/plans/2026-05-31-robotics-behavior-collaboration/` (THIS) — plan in 4 formats.
- Already shipped: `signedkan_wip/src/vision/fuzzy_pose.py` (FuzzySignature / HSiKAN / HSiKAN-Gömb pose models + soft-argmax + synthetic dataset).
- Stage 2: `signedkan_wip/src/vision/pose_real.py` — MPII / COCO loader + smoke harness.
- Stage 3: `signedkan_wip/src/temporal/fuzzy_behavior.py` — 1-D temporal FuzzySignatureLayer + behavior head.
- Stage 4: `signedkan_wip/src/multiagent/coop_graph.py` — streaming signed-graph wrapper for multi-agent observation.

CORE.YAML items: none touched. signedkan_wip/ is research-only.

Performance budget (per stage)

Stage	Wall budget	Memory
1 — synthetic pose smoke (3 models, paired)	≤ 5 h	≤ 4 GiB GPU
2 — real pose (MPII subset)	≤ 1 day	≤ 8 GiB GPU
3 — behavior recognition (NTU-60)	≤ 3 days	≤ 8 GiB GPU
4 — multi-agent sim (2-D toy world)	≤ 2 days	CPU-feasible

All within CLAUDE.md §4’s 16 GiB RSS cap.

Risk anticipation

- **Stage 1 gate failure:** addressed in the stage-1 plan (2026-05-31-fuzzy-pose-detection/plan.tex).
- **Stage 2 dataset overfit:** at sub-10k params the model is in a regime where the dataset capacity matters less than the inductive bias. Risk: the fuzzy / Gömb inductive bias may not align with raw RGB pose data. Mitigation: use coarse-grained depth or skeleton heatmaps as input instead of raw RGB.
- **Stage 3 temporal fuzzy collapse:** the 18-cell bounded-fuzzy failure (from the rev-2 to rev-4 path in the classification work) might recur in temporal aggregation if we again use only bounded $[0, 1]$ -valued internals. Mitigation: re-use the validated rev-5 “additive_centered” residual from day one.
- **Stage 4 sim-to-real gap:** simulation environments don’t capture real robot sensor noise. Mitigation: defer real-robot deployment to a post-plan iteration; this plan ends at the simulated validation.

Rollback path

Each stage is independently revertible: deleting a stage’s added files and tests reverts to the prior stage’s validated state. The base perception layer (Stage 1) is already isolated from the existing rev-6 MNIST work.

Empty-plan-dir hygiene

If the plan is abandoned before stage 1’s smoke completes, delete this plan dir.